



## Assignment #2 (Data Encryption Standard DES)

ID: \_\_\_\_\_

**NOTE :**

- Q.1 Derive the 32-bit output for the 48-bit input using the S-Box table given in the lecture slides:

|   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |   |
|---|---|---|---|---|---|---|---|---|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 |   |
| 0 | 1 | 1 | 0 | 1 | 0 | 0 | 1 | 1 | 1  | 1  | 0  | 0  | 1  | 0  | 1  | 0  | 1  | 0  | 1  | 0  | 0  | 0  | 0  | 1 |

|    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 |
| 1  | 1  | 0  | 1  | 1  | 1  | 0  | 0  | 0  | 1  | 0  | 0  | 1  | 0  | 1  | 0  | 0  | 1  | 1  | 1  | 0  | 0  | 0  | 1  |

|          | 6-bit input | Row value | Column value | S-box result | 4-bit output |
|----------|-------------|-----------|--------------|--------------|--------------|
| S-Box S1 |             |           |              |              |              |
| S-Box S2 |             |           |              |              |              |
| S-Box S3 |             |           |              |              |              |
| S-Box S4 |             |           |              |              |              |
| S-Box S5 |             |           |              |              |              |
| S-Box S6 |             |           |              |              |              |
| S-Box S7 |             |           |              |              |              |
| S-Box S8 |             |           |              |              |              |

[illegible][illegible]

Q2. Consider the 48-bit input given in Question 1. Invert the leftmost bit of each of the eight 6-bit input blocks. By “invert”, what I mean is that if the leftmost bit is a 0, change it to 1 and if the leftmost bit is a 1, change it to 0.

a) Determine the 4-bit output from each of the eight S-boxes and find your 32-bit output.

b) Determine what fraction of your 32-bit output is different from the 32-bit output obtained in Question 1.

**Solution:**

Q3. What is the output of the first round of the DES algorithm when the plaintext and the key both are zero?

**Solution:**

Q4. Initial key halves are given in DES use lecture slides for required tables and generate round keys for round 1 and 2.

C0 = 1010101010101010101010101010

D0 = 0101010101010101010101010101

**Solution:**

